

Mathematics Questions - 3d geometry

Subject: Mathematics | Chapter: 3d geometry | Difficulty: hard

Question 1: Find the equation of the plane passing through the points A(1, 2, 3), B(2, 3, 1), and C(3, 1, 2).

Options:

- A) $x + y + z = 6$
- B) $x - y + z = 0$
- C) $x + y - z = 0$
- D) $x - y - z = -4$

Difficulty: 8/5

Question 2: Determine the shortest distance between the lines $(x-1)/2 = (y-2)/3 = (z-3)/4$ and $(x-2)/3 = (y-4)/4 = (z-5)/5$.

Options:

- A) $1/\sqrt{6}$
- B) 1
- C) $\sqrt{6}$
- D) 0

Difficulty: 9/5

Question 3: A plane passes through (1, 2, 3) and is perpendicular to the line joining (2, 3, 4) and (3, 4, 5). Find its equation.

Options:

- A) $x + y + z = 6$
- B) $x + y + z = 1$
- C) $x + y + z = 0$
- D) $x + y + z = 7$

Difficulty: 7/5

Question 4: Find the angle between the planes $2x + y - z = 5$ and $x + 2y + z = 10$.

Options:

- A) $\cos^{-1}(1/6)$
- B) $\cos^{-1}(1/2)$
- C) $\cos^{-1}(1/3)$
- D) $\cos^{-1}(1/\sqrt{6})$

Difficulty: 7/5

Question 5: If a line makes angles α, β, γ with the axes, then find the value of $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma$.

Options:

- A) 1
- B) 2
- C) 0
- D) -1

Difficulty: 6/5

Question 6: Find the image of the point (1, 3, 4) in the plane $2x - y + z + 3 = 0$.

Options:

- A) (-3, 5, 2)
- B) (3, -5, -2)
- C) (-3, -5, 2)
- D) (3, 5, 2)

Difficulty: 9/5

Question 7: Find the equation of the plane that passes through the point (1, 1, 1) and is parallel to the plane $2x + 3y - z = 4$.

Options:

- A) $2x + 3y - z = 4$
- B) $2x + 3y - z = 0$
- C) $2x + 3y - z = 6$
- D) $2x + 3y - z = -4$

Difficulty: 6/5

Question 8: Two planes intersect at an angle of $\frac{\pi}{3}$. If the equation of one plane is $x + y + z = 1$, find a possible equation for the second plane.

Options:

- A) $x - y - z = 1$
- B) $x + y - z = 2$
- C) $x - y + z = 0$
- D) $2x + y + 2z = 3$

Difficulty: 10/5

Question 9: A line passes through (1, 2, 3) and is parallel to the vector $\langle 2, -1, 3 \rangle$. Find the coordinates of the point where the line intersects the plane $x + y + z = 10$.

Options:

- A) (3, 1, 6)
- B) (4, 0, 6)
- C) (2, 3, 5)
- D) (0, 4, 6)

Difficulty: 8/5

Question 10: A rectangular box has vertices at $(0,0,0)$, $(a,0,0)$, $(0,b,0)$, $(0,0,c)$, and others. Find the angle between the diagonals of the box.

Options:

- A) $\cos \{^1((a^2 + b^2 + c^2) / (a^2 + b^2 + c^2)^{1/2})$
- B) $\cos \{^1((a^2 - b^2 - c^2) / (a^2 + b^2 + c^2))$
- C) $\cos \{^1((a^2 + b^2 - c^2) / (a^2 + b^2 + c^2))$
- D) $\cos \{^1((a^2 + b^2 + c^2) / (a^2 + b^2 + c^2)^{1/2})$

Difficulty: 9/5

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